

Linux Class 5 - Practice Task

Practice of wc, head, tail, pipe and tee

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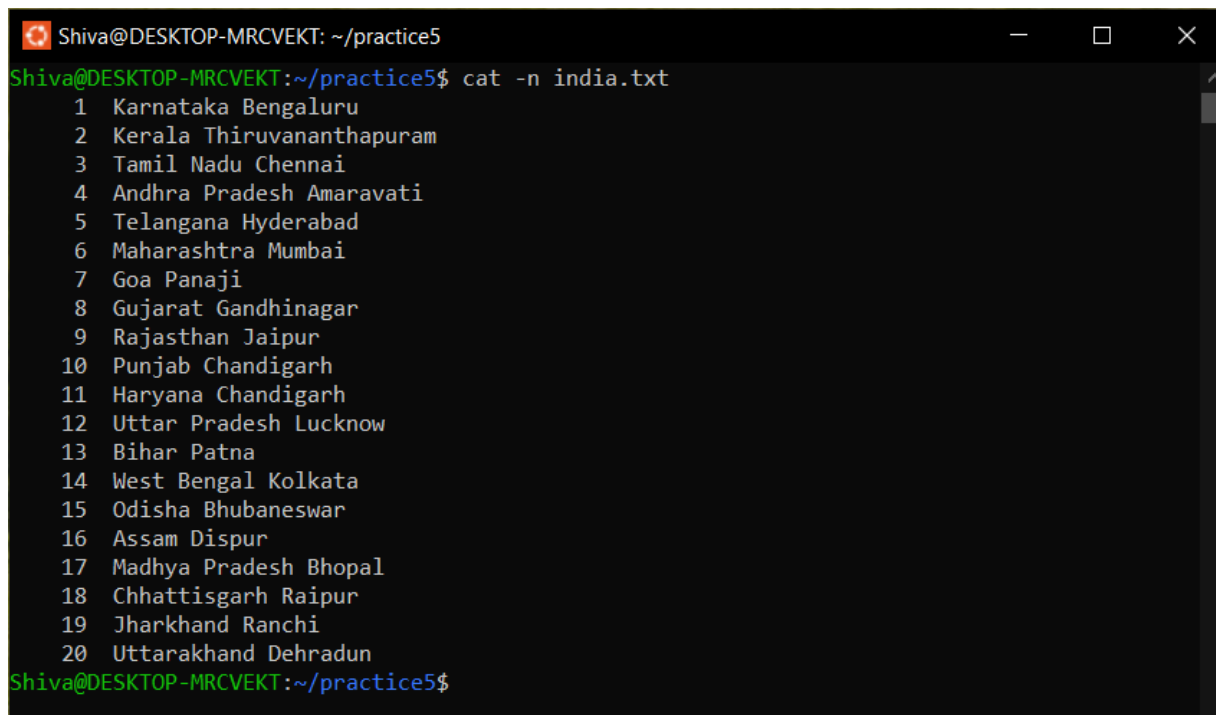
Date: 26 September 2026

Practice for class 5. I made one folder called practice5 and did all the commands from the notes one by one.

My practice files

For head and tail I need a file with many lines, so I made india.txt with 20 states of India and their capitals, one state in each line. I typed it using `cat > india.txt` and pressed Ctrl+D at the end to save it. I also made one small file cricket.txt with 3 lines about Virat, Dhoni and Rohit.

```
cat -n india.txt
```

A screenshot of a terminal window titled 'Shiva@DESKTOP-MRCVEKT: ~/practice5'. The prompt is 'Shiva@DESKTOP-MRCVEKT:~/practice5\$'. The command 'cat -n india.txt' has been executed, displaying a list of 20 Indian states and their capitals, each preceded by a line number from 1 to 20. The list is as follows:
1 Karnataka Bengaluru
2 Kerala Thiruvananthapuram
3 Tamil Nadu Chennai
4 Andhra Pradesh Amaravati
5 Telangana Hyderabad
6 Maharashtra Mumbai
7 Goa Panaji
8 Gujarat Gandhinagar
9 Rajasthan Jaipur
10 Punjab Chandigarh
11 Haryana Chandigarh
12 Uttar Pradesh Lucknow
13 Bihar Patna
14 West Bengal Kolkata
15 Odisha Bhubaneswar
16 Assam Dispur
17 Madhya Pradesh Bhopal
18 Chhattisgarh Raipur
19 Jharkhand Ranchi
20 Uttarakhand Dehradun
The prompt 'Shiva@DESKTOP-MRCVEKT:~/practice5\$' is visible at the bottom of the terminal output.

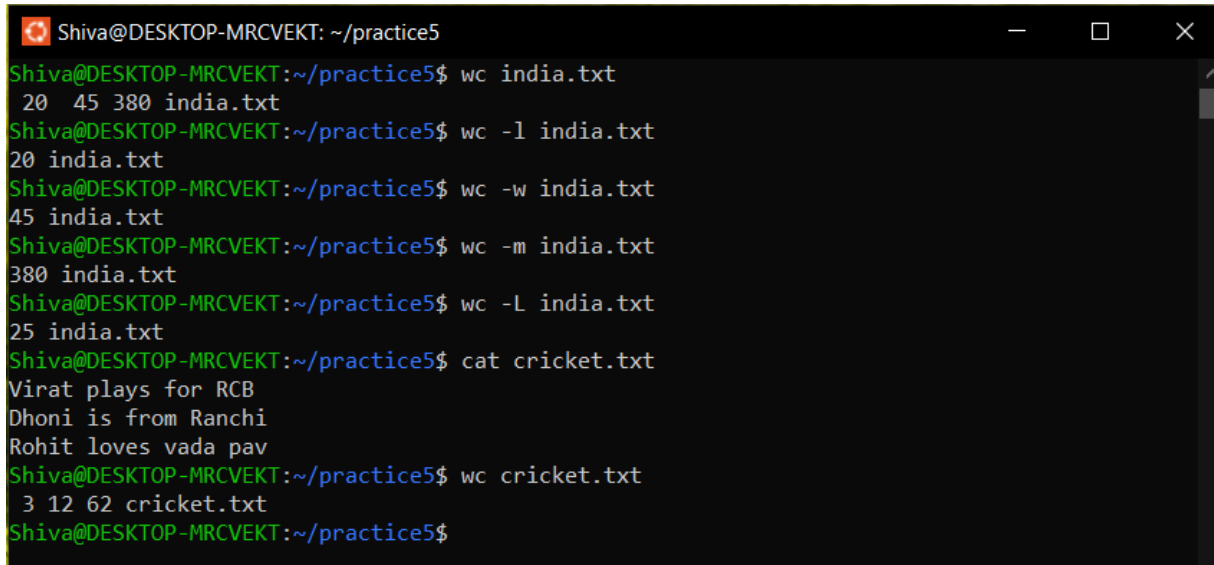
Screenshot 1: my india.txt file with line numbers.

I used `cat -n` so that I can see the line numbers. This helped me a lot later to check if head and tail are giving correct lines or not.

wc command (word count)

wc counts the lines, words and characters in a file.

```
wc india.txt
wc -l india.txt
wc -w india.txt
wc -m india.txt
wc -L india.txt
wc cricket.txt
```



```
Shiva@DESKTOP-MRCVEKT: ~/practice5
Shiva@DESKTOP-MRCVEKT:~/practice5$ wc india.txt
 20  45 380 india.txt
Shiva@DESKTOP-MRCVEKT:~/practice5$ wc -l india.txt
20 india.txt
Shiva@DESKTOP-MRCVEKT:~/practice5$ wc -w india.txt
45 india.txt
Shiva@DESKTOP-MRCVEKT:~/practice5$ wc -m india.txt
380 india.txt
Shiva@DESKTOP-MRCVEKT:~/practice5$ wc -L india.txt
25 india.txt
Shiva@DESKTOP-MRCVEKT:~/practice5$ cat cricket.txt
Virat plays for RCB
Dhoni is from Ranchi
Rohit loves vada pav
Shiva@DESKTOP-MRCVEKT:~/practice5$ wc cricket.txt
 3 12 62 cricket.txt
Shiva@DESKTOP-MRCVEKT:~/practice5$
```

Screenshot 2: wc with all the options.

Plain wc gave 3 numbers together: 20 45 380. First one is lines, second is words and third is characters. With options it gives only one number.

Command	Answer	What it means
wc -l india.txt	20	20 lines
wc -w india.txt	45	45 words
wc -m india.txt	380	380 characters (spaces and new line also counted)
wc -L india.txt	25	longest line has 25 characters

At first I thought 45 words is wrong, because there are 20 states. Then I understood that names like Tamil Nadu and Uttar Pradesh have 2 words, so total becomes 45. And -L gave 25 because Kerala Thiruvananthapuram is the longest line. Thiruvananthapuram is a very long name!

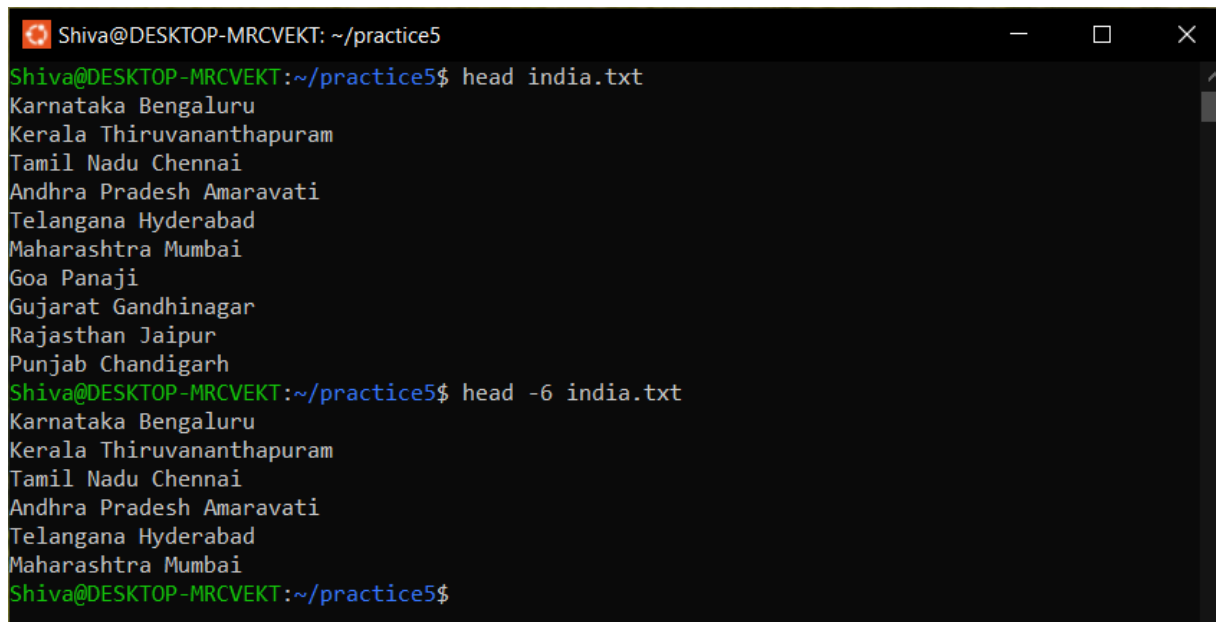
For cricket.txt it gave 3 lines, 12 words and 62 characters.

head command

head shows the first lines of a file. If I do not give any number, it shows the first 10 lines.

```
head india.txt
```

```
head -6 india.txt
```

A terminal window titled 'Shiva@DESKTOP-MRCVEKT: ~/practice5' showing the output of the 'head' command. The first command 'head india.txt' displays the first 10 lines of the file: Karnataka Bengaluru, Kerala Thiruvananthapuram, Tamil Nadu Chennai, Andhra Pradesh Amaravati, Telangana Hyderabad, Maharashtra Mumbai, Goa Panaji, Gujarat Gandhinagar, Rajasthan Jaipur, and Punjab Chandigarh. The second command 'head -6 india.txt' displays the first 6 lines: Karnataka Bengaluru, Kerala Thiruvananthapuram, Tamil Nadu Chennai, Andhra Pradesh Amaravati, Telangana Hyderabad, and Maharashtra Mumbai.

```
Shiva@DESKTOP-MRCVEKT: ~/practice5
Shiva@DESKTOP-MRCVEKT:~/practice5$ head india.txt
Karnataka Bengaluru
Kerala Thiruvananthapuram
Tamil Nadu Chennai
Andhra Pradesh Amaravati
Telangana Hyderabad
Maharashtra Mumbai
Goa Panaji
Gujarat Gandhinagar
Rajasthan Jaipur
Punjab Chandigarh
Shiva@DESKTOP-MRCVEKT:~/practice5$ head -6 india.txt
Karnataka Bengaluru
Kerala Thiruvananthapuram
Tamil Nadu Chennai
Andhra Pradesh Amaravati
Telangana Hyderabad
Maharashtra Mumbai
Shiva@DESKTOP-MRCVEKT:~/practice5$
```

Screenshot 3: head shows first 10 lines, head -6 shows first 6 lines.

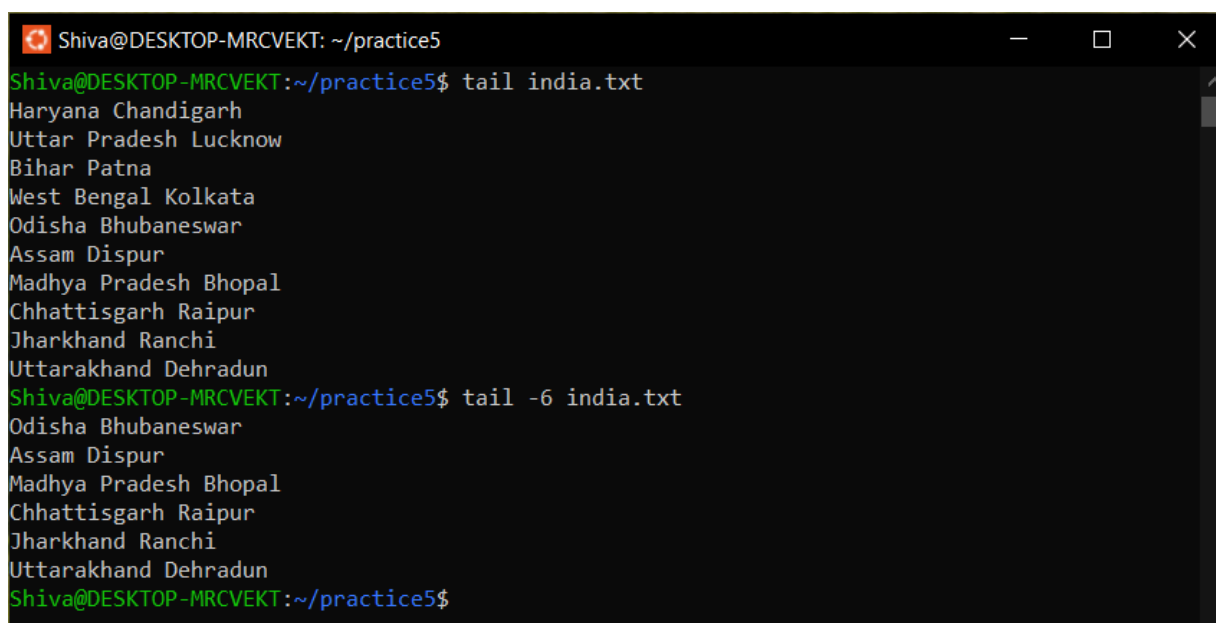
head india.txt stopped at Punjab, which is line 10. head -6 stopped at Maharashtra, which is line 6. So the number after - is how many lines we want.

tail command

tail is the opposite of head. It shows the last lines of the file. Default is also 10 lines.

```
tail india.txt
```

```
tail -6 india.txt
```

A terminal window titled 'Shiva@DESKTOP-MRCVEKT: ~/practice5' showing the output of the 'tail' command. The first command 'tail india.txt' displays the last 10 lines of the file: Haryana Chandigarh, Uttar Pradesh Lucknow, Bihar Patna, West Bengal Kolkata, Odisha Bhubaneswar, Assam Dispur, Madhya Pradesh Bhopal, Chhattisgarh Raipur, Jharkhand Ranchi, and Uttarakhand Dehradun. The second command 'tail -6 india.txt' displays the last 6 lines: Odisha Bhubaneswar, Assam Dispur, Madhya Pradesh Bhopal, Chhattisgarh Raipur, Jharkhand Ranchi, and Uttarakhand Dehradun.

```
Shiva@DESKTOP-MRCVEKT: ~/practice5
Shiva@DESKTOP-MRCVEKT:~/practice5$ tail india.txt
Haryana Chandigarh
Uttar Pradesh Lucknow
Bihar Patna
West Bengal Kolkata
Odisha Bhubaneswar
Assam Dispur
Madhya Pradesh Bhopal
Chhattisgarh Raipur
Jharkhand Ranchi
Uttarakhand Dehradun
Shiva@DESKTOP-MRCVEKT:~/practice5$ tail -6 india.txt
Odisha Bhubaneswar
Assam Dispur
Madhya Pradesh Bhopal
Chhattisgarh Raipur
Jharkhand Ranchi
Uttarakhand Dehradun
Shiva@DESKTOP-MRCVEKT:~/practice5$
```

Screenshot 4: tail shows last 10 lines, tail -6 shows last 6 lines.

tail india.txt started from Haryana (line 11) and went till Uttarakhand (line 20). tail -6 showed only from Odisha to Uttarakhand, that is lines 15 to 20.

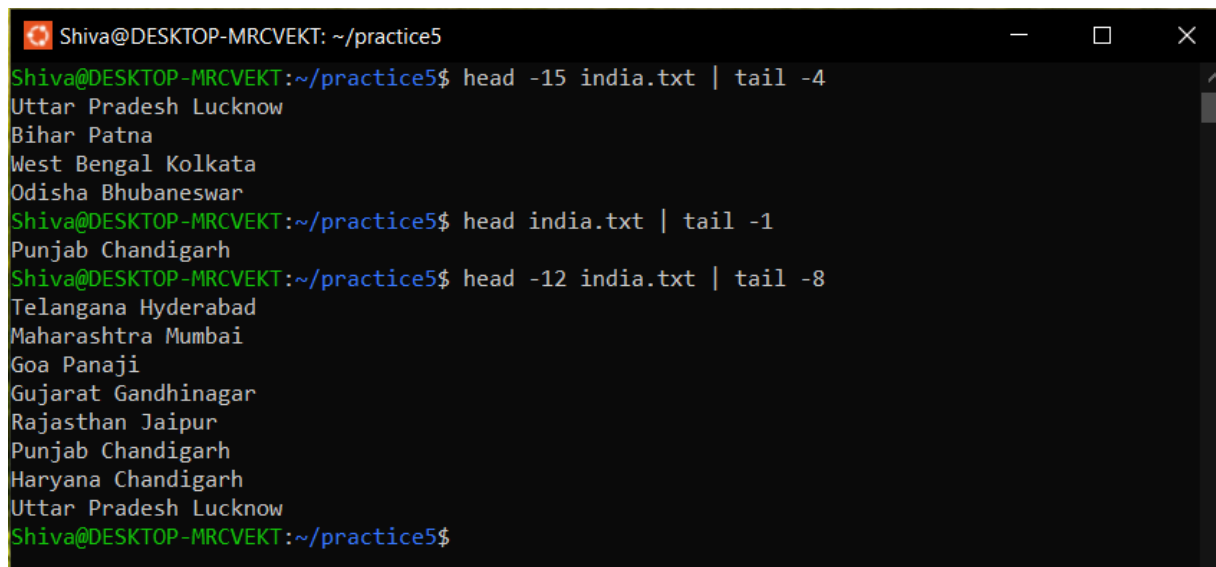
Pipe operator (|)

The pipe sends the output of the first command as input to the second command. Using head and tail together with a pipe, I can take any lines from the middle of the file. This part was little confusing for me in class, so I tried all the examples from the notes.

```
head -15 india.txt | tail -4
```

```
head india.txt | tail -1
```

```
head -12 india.txt | tail -8
```



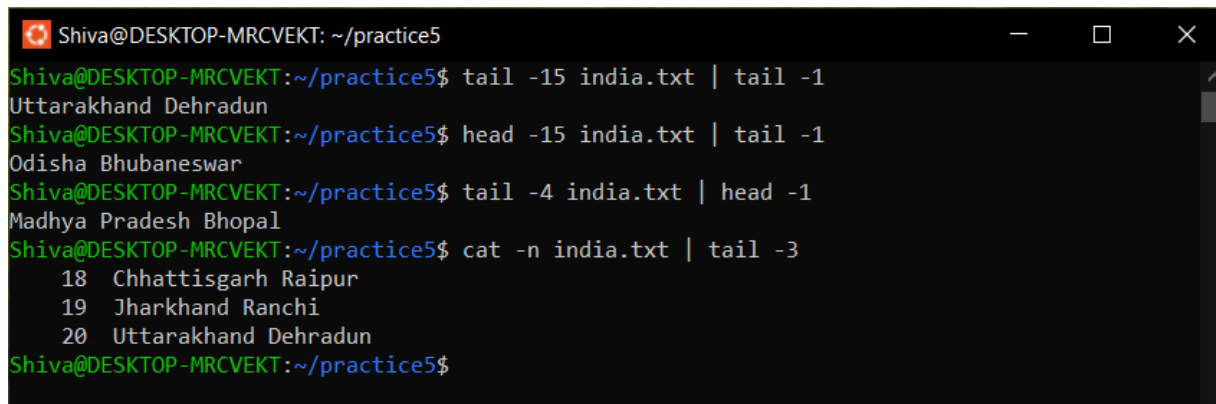
```
Shiva@DESKTOP-MRCVEKT: ~/practice5
Shiva@DESKTOP-MRCVEKT:~/practice5$ head -15 india.txt | tail -4
Uttar Pradesh Lucknow
Bihar Patna
West Bengal Kolkata
Odisha Bhubaneswar
Shiva@DESKTOP-MRCVEKT:~/practice5$ head india.txt | tail -1
Punjab Chandigarh
Shiva@DESKTOP-MRCVEKT:~/practice5$ head -12 india.txt | tail -8
Telangana Hyderabad
Maharashtra Mumbai
Goa Panaji
Gujarat Gandhinagar
Rajasthan Jaipur
Punjab Chandigarh
Haryana Chandigarh
Uttar Pradesh Lucknow
Shiva@DESKTOP-MRCVEKT:~/practice5$
```

Screenshot 5: taking lines from the middle using head and tail together.

- head -15 | tail -4: head first takes lines 1 to 15, then tail takes the last 4 of those. So I got lines 12 to 15 (Uttar Pradesh to Odisha).
- head | tail -1: head takes the first 10 lines and tail -1 takes the last one of them. So only the 10th line came, Punjab Chandigarh.
- head -12 | tail -8: first 12 lines, then last 8 of them, so lines 5 to 12 (Telangana to Uttar Pradesh).

The trick I understood is: the head number is the last line I want, and the tail number is how many lines I want before it.

```
tail -15 india.txt | tail -1
head -15 india.txt | tail -1
tail -4 india.txt | head -1
cat -n india.txt | tail -3
```



```
Shiva@DESKTOP-MRCVEKT: ~/practice5
Shiva@DESKTOP-MRCVEKT:~/practice5$ tail -15 india.txt | tail -1
Uttarakhand Dehradun
Shiva@DESKTOP-MRCVEKT:~/practice5$ head -15 india.txt | tail -1
Odisha Bhubaneswar
Shiva@DESKTOP-MRCVEKT:~/practice5$ tail -4 india.txt | head -1
Madhya Pradesh Bhopal
Shiva@DESKTOP-MRCVEKT:~/practice5$ cat -n india.txt | tail -3
18 Chhattisgarh Raipur
19 Jharkhand Ranchi
20 Uttarakhand Dehradun
Shiva@DESKTOP-MRCVEKT:~/practice5$
```

Screenshot 6: the other pipe examples from the notes.

In the notes it is written that `tail -15 | tail -1` shows the 15th line. But when I tried, it showed Uttarakhand Dehradun, which is the last line (line 20). I think it is because `tail -15` gives lines 6 to 20, and `tail -1` again takes the last one, so it is always the last line of the file. To get the 15th line I used `head -15 | tail -1` and it gave Odisha Bhubaneswar, which is correct. I checked with my `cat -n` screenshot.

`tail -4 | head -1` gave Madhya Pradesh Bhopal. The last 4 lines are 17 to 20, and `head -1` takes the first one of them, so it is line 17. Correct.

`cat -n | tail -3` showed the last 3 lines with their line numbers 18, 19 and 20. This is useful when we want to know the line number also.

tee command

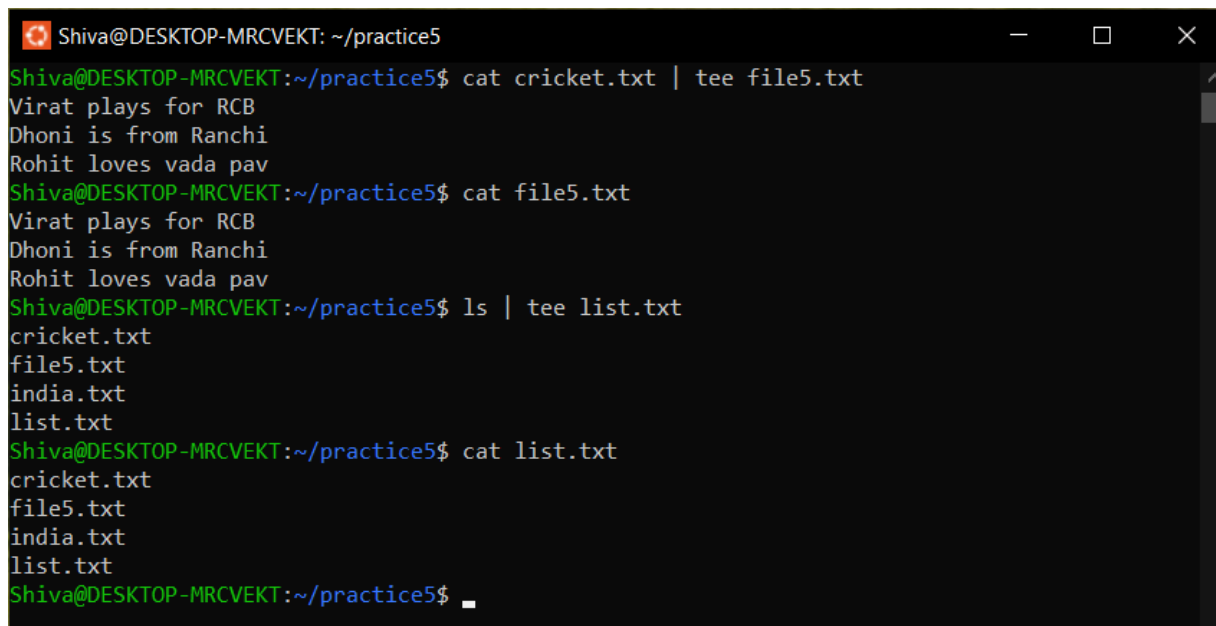
tee is used with pipe. It shows the output on the screen and at the same time saves it into a file. I think the name is from the T shape pipe used in plumbing, one input and two outputs.

```
cat cricket.txt | tee file5.txt
```

```
cat file5.txt
```

```
ls | tee list.txt
```

```
cat list.txt
```

A screenshot of a terminal window titled 'Shiva@DESKTOP-MRCVEKT: ~/practice5'. The terminal shows a series of commands and their outputs. First, 'cat cricket.txt | tee file5.txt' is executed, showing three lines of text: 'Virat plays for RCB', 'Dhoni is from Ranchi', and 'Rohit loves vada pav'. Then, 'cat file5.txt' is executed, displaying the same three lines. Next, 'ls | tee list.txt' is executed, showing a list of files: 'cricket.txt', 'file5.txt', 'india.txt', and 'list.txt'. Finally, 'cat list.txt' is executed, displaying the same list of files. The prompt 'Shiva@DESKTOP-MRCVEKT:~/practice5\$' is visible at the end of each command line.

```
Shiva@DESKTOP-MRCVEKT:~/practice5$ cat cricket.txt | tee file5.txt
Virat plays for RCB
Dhoni is from Ranchi
Rohit loves vada pav
Shiva@DESKTOP-MRCVEKT:~/practice5$ cat file5.txt
Virat plays for RCB
Dhoni is from Ranchi
Rohit loves vada pav
Shiva@DESKTOP-MRCVEKT:~/practice5$ ls | tee list.txt
cricket.txt
file5.txt
india.txt
list.txt
Shiva@DESKTOP-MRCVEKT:~/practice5$ cat list.txt
cricket.txt
file5.txt
india.txt
list.txt
Shiva@DESKTOP-MRCVEKT:~/practice5$
```

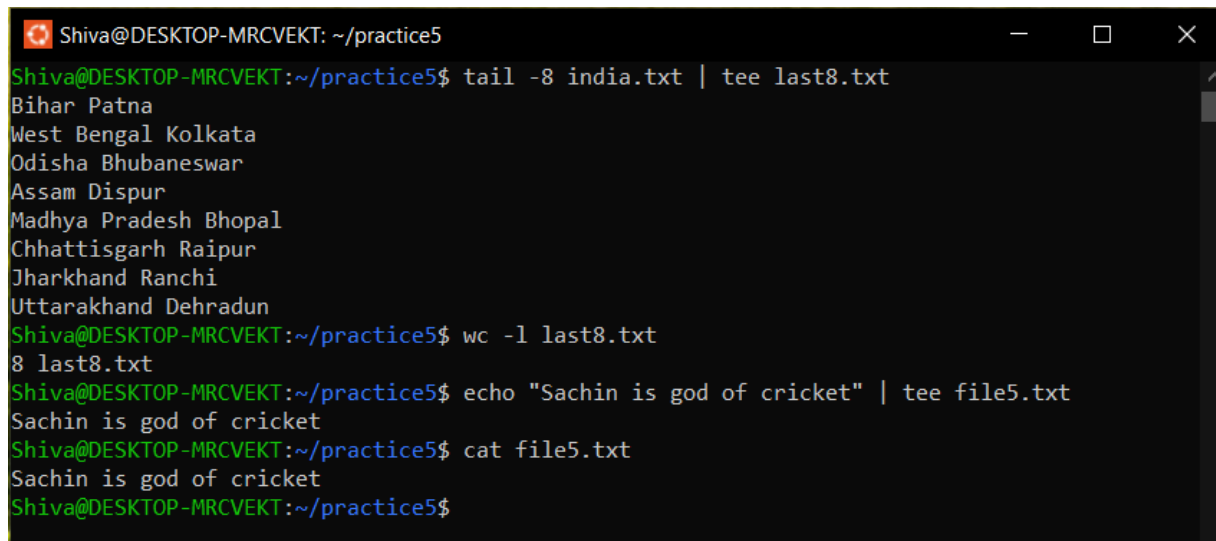
Screenshot 7: tee shows the output and also saves it in a file.

The 3 cricket lines came on the screen, and when I did cat file5.txt the same 3 lines were there inside the file. So tee did both things in one go.

ls | tee list.txt saved the list of my files into list.txt. One funny thing, list.txt itself is also there in the list. I think tee creates the file first and then ls runs, so ls can see it.

For the third example I used my own file name, last8.txt. I also checked the point in the notes that tee replaces the old content of the file.

```
tail -8 india.txt | tee last8.txt
wc -l last8.txt
echo "Sachin is god of cricket" | tee file5.txt
cat file5.txt
```

A screenshot of a terminal window titled 'Shiva@DESKTOP-MRCVEKT: ~/practice5'. The terminal shows the following commands and their outputs: 1. 'tail -8 india.txt | tee last8.txt' outputs eight Indian states: Bihar Patna, West Bengal Kolkata, Odisha Bhubaneswar, Assam Dispur, Madhya Pradesh Bhopal, Chhattisgarh Raipur, Jharkhand Ranchi, and Uttarakhand Dehradun. 2. 'wc -l last8.txt' outputs '8 last8.txt'. 3. 'echo "Sachin is god of cricket" | tee file5.txt' outputs 'Sachin is god of cricket'. 4. 'cat file5.txt' outputs 'Sachin is god of cricket'. The prompt 'Shiva@DESKTOP-MRCVEKT:~/practice5\$' is visible at the end of each command line.

```
Shiva@DESKTOP-MRCVEKT: ~/practice5
Shiva@DESKTOP-MRCVEKT:~/practice5$ tail -8 india.txt | tee last8.txt
Bihar Patna
West Bengal Kolkata
Odisha Bhubaneswar
Assam Dispur
Madhya Pradesh Bhopal
Chhattisgarh Raipur
Jharkhand Ranchi
Uttarakhand Dehradun
Shiva@DESKTOP-MRCVEKT:~/practice5$ wc -l last8.txt
8 last8.txt
Shiva@DESKTOP-MRCVEKT:~/practice5$ echo "Sachin is god of cricket" | tee file5.txt
Sachin is god of cricket
Shiva@DESKTOP-MRCVEKT:~/practice5$ cat file5.txt
Sachin is god of cricket
Shiva@DESKTOP-MRCVEKT:~/practice5$
```

Screenshot 8: copying the last 8 lines, and tee replacing the old content.

tail -8 | tee last8.txt showed the last 8 states and saved them, and wc -l last8.txt says 8, so it is correct.

Then I sent one line about Sachin into file5.txt using tee. Before this, file5.txt had the 3 cricket lines. After this, cat file5.txt shows only the Sachin line. So yes, tee replaces the full content of the file, same as the notes say. If we want to add at the end without deleting, I read that tee -a is there for that.